ModelIt!

Student Activity Pack

The Rock That Remembers Everything

How Earth's Rocks Tell the Story of Millions of Years

Grade 6 | NGSS MS-ESS2-1, MS-ESS2-3 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where do you think rocks come from? Have they always existed?

2 How do you think a seashell could end up inside a rock on a mountain?

3 Do you think a rock can change into a different kind of rock? How?

4 Draw what you think happens to a rock over millions of years.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Heat & Pressure | Weathering Rate | Sediment Layers | Time

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Mountain Builder

Maximum heat and pressure, low weathering

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Erosion Machine

Maximum weathering, moderate other factors

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Deep Time

All moderate, 500 million year timeframe

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Volcanic Activity — Eruptions produce new igneous rock from cooled magma and lava

Water Erosion — Moving water is the most powerful weathering agent, carving canyons and transporting sediment

Biological Weathering — Plant roots, burrowing animals, and microorganisms break down rock chemically and physically

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a rock on top of a mountain contain seashells from the bottom of an ancient ocean? Use evidence from your simulation to support your answers.

1 Why are fossils only found in sedimentary rock and not in igneous or metamorphic rock?

2 If the atoms in a rock have been recycled for billions of years, what were they part of before?

3 How is the rock cycle connected to plate tectonics?

4 Why is the rock cycle called a 'cycle' if it doesn't always happen in the same order?

5 How does studying rocks help us understand what Earth looked like millions of years ago?

8 STEM Challenge: Design a Geological Time Capsule

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YOUR ENGINEERING MISSION

Design a system to preserve a message for 1 million years using your understanding of how rocks form, weather, and preserve materials.

THE SCENARIO

A science foundation wants to create a time capsule that will survive for 1 million years. Your geology team must use evidence from the rock cycle model to recommend the best materials, location, and protection strategy.

Engineering Design Thinking

1. WHAT MATERIAL WILL YOU ENCASE YOUR MESSAGE IN — AND WHAT ROCK PROPERTIES MAKE IT DURABLE?
2. WHERE ON EARTH WOULD YOU PLACE YOUR TIME CAPSULE TO MINIMIZE WEATHERING AND TECTONIC ACTIVITY?
3. HOW WILL YOU PROTECT AGAINST HEAT, PRESSURE, WATER, AND BIOLOGICAL DEGRADATION?
4. HOW WILL FUTURE DISCOVERERS KNOW WHAT THEY'VE FOUND AND HOW TO READ YOUR MESSAGE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Geologists and Paleontologists study rocks, fossils, and Earth's history to find natural resources, predict hazards, and understand our planet's past. They earn $60,000–$115,000/year.